merals) in such a setting — this finding is consistent with work that has shown human solvers also find a problem to be more difficult when the operator word is explicit but unfamiliar (Derzhanski and Veneva, 2018). Overall, our results demonstrate that it is difficult for models to reason about the abstract idea that linguistic quantities might contain *operators*, if the operators are not explicitly provided using familiar symbols.

4.1.1 Error analysis

We observe some common patterns of error in the model responses. For the three problems which involved squares and cubes of numbers, when the operators were not explicit and familiar, o1-mini almost always responded by pattern-matching (e.g. providing another square/cube number) instead of solving the problem, as seen in Table 2. o1-mini also reproduced a number given in the input question as the answer in several cases (11 for the multi-token condition, and 3 for the single-token condition, across 150 trials) when the operators were not explicit and familiar.

Condition	Single	Multi
explicit symbol	8	4
explicit random word	0	8
implicit	14	5

Table 2: Incorrect pattern-matched square / cube answers (out of 15 possible trials)

Further, when o1-mini answered a problem incorrectly, its responses were often inconsistent across the five trials of that problem. Notably, in 50% of single-character cases lacking explicit and familiar operators, all five responses were distinct and incorrect. This further shows that performance appears to depend on the presence of explicit operator cues; in their absence, o1-mini does not reliably solve the problem.

4.2 Providing contextual information

Our first experiment showed that in the absence of problem-specific instructions, when given a linguistic-mathematical problem directly, LLMs struggle to solve it unless the operations are both explicit and familiar. This leaves open the question of whether providing additional problem-specific information would affect the model performance. We thus modulate the context of the problem in three different ways. We query the same four problem variants as described in Table 1, additionally providing the following contextual information:

Language: "Here is a puzzle *based on numbers in the {language} language*."

Base: "Here is a puzzle *based on numbers in a language that uses a base-{n} numeral system*."

Implicit operations: "Here is a puzzle *based on numbers in a language. In this language, numbers may be constructed through implicit operations like addition (twenty-nine = 20 + 9) or multiplication (five hundred = 5 × 100)*." [only for IMPLICIT condition]

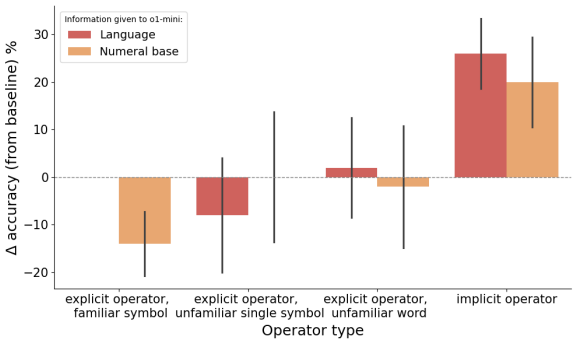

Figure 2: **Language and base information only helps in the IMPLICIT case.** Effect of adding language or numeral base information, plotted as a difference from the baseline values in Figure 1 for o1-mini. In cases with explicit operators, conflating overtly mathematical and linguistic information appears to confuse the models.

We compare these to the baseline results from Section 4.1 for o1-mini, presenting our results in Figure 2. In cases *other* than the implicit operator condition, the model seems to recognize the problem as requiring a more mathematical kind of reasoning, so providing linguistic information seems to confuse the model and average performance is worse. However, in the implicit operator (A B) condition, model performance improves significantly, perhaps because the setting of the problem is less overtly mathematical. In Figure 3, we show that providing information about the implicit reasoning needed is not as significant a boost as activating knowledge about the specific language.

4.3 Ablations: constructed minimal-pair problems

In order to ensure that it is the difference in operators (as opposed to other features of the numeral system) that explains the models' inability to solve these problems, we performed an ablation study to test whether models could handle other aspects

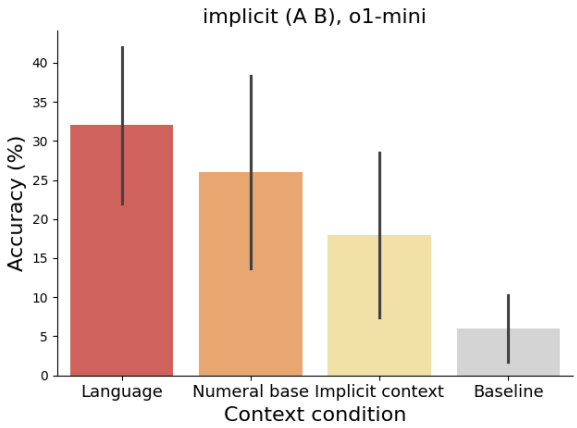

Figure 3: **Extra information improves performance on IMPLICIT problems (A B)**. Information about implicitness is helpful, but not as much as more direct information like the problem language. Error bars denote standard error of the mean. 5 iterations / problem.

Parameter	GPT-3.5	GPT-4	o1-mini
Numeral system vs. not AB = fifty one | AB = big bird	✓	✓	✓
Typed vs. glyph AB = fifty one | AB = 51	✓	✓	✓
Order L → R vs. L ← R AB = 51 | BA = 51	✓	✓	✓
Additive vs. subtractive AB = 27 (20 + 7) | AB = 27 (30 - 3)	✗	✓	✓
Base of the numeral system*	✗	✓	✓

Table 3: Minimal pair results: GPT-4 and o1-mini solve all problem pairs, GPT-3.5-turbo struggles with numeral base and combination. DeepSeek-R1-distill-Qwen-7B does not produce an answer in the right format for most settings, so it is excluded from this table. Further data on testing all bases 4-19 linked in Table 6.

of numeral construction and combination. Our experiment is inspired by the notion of a linguistic *minimal pair*, a pair of linguistic items that differ in exactly one meaningful element. We construct minimal pairs of simple, synthetic number system problems, where every element is the same except for one specific parameter that differs between two paired problems. We tested five major parameters of numeral systems, as described in Table 3.

L → R	L ← R
A B = 51	B A = 51
A C = 57	C A = 57
D B = 41	B D = 41
D C = ??	C D = ??

Figure 4: Example of full minimal pair template problem, for the Order parameter, where we varied whether digits are read left-to-right or right-to-left.

In all cases, GPT-4 and more advanced models could solve the template problems. It thus appears that most basic "building blocks" of number systems (e.g. the base of the system, the order of numerals, etc.) did not affect model performance in isolation, but the models consistently fail to solve number problems that involve constructing and combining complex numerals.

5 Discussion and Conclusions

We study the entanglement between linguistic and numeric knowledge in language models, focusing on the ability of models to use mathematical reasoning in problems that display the implicit numerical structure in language. In the setting of these linguistic-mathematical puzzles, we show that the overtness and familiarity of operators affects the performance of language models, although many humans are able to understand how numeral systems work and hence solve the problems without needing specified operators. However, a broader study with different controls and parameter settings remains open for future work. Since all our evaluation was standardized and closed-form, we welcome research on open-ended evaluation of reasoning task responses. Current language models seem to display some level of emergent modular structure (Teehan et al., 2022; Lepori et al., 2023) — perhaps linguistic and mathematical tasks activate separate circuits or subspaces in models, and understanding the ways in which reasoning fine-tuning and reinforcement learning interacts with linguistic pretraining is another promising avenue for future research. Investigating such questions enriches our understanding of both computational and human approaches to representing numbers in language. The ability to understand language and abstract rule-governed systems is a fascinating aspect of human intelligence, and we hope that our research provides some insight into the understanding of this remarkable human trait.

Limitations

We acknowledge the possibility that our results are explained by limitations in the training data and the small size of our dataset, as language models often equal human performance on benchmarks for which they have large quantities of similar-enough training data (Achiam et al., 2023). Perhaps an LLM trained on a massive corpus of linguistic number system problems would be able to solve new, previously unseen number system problems. But the data today are far too limited for such an approach, and crucially, a human solver who is familiar with existing number system problems can generalize to unseen problems extremely well! Even a human solver who is unfamiliar with existing number system problems can in theory solve any problem they are provided just by logically reasoning. Importantly, we note that although this may not be true of the *average* human, when comparing the top end of humans with the top-performing current language models, it is clear that intuiting rules from human-scale data is still challenging for LLMs.

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References

Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat, and 1 others. 2023. GPT-4 Technical Report. *arXiv preprint arXiv:2303.08774*.

Janice Ahn, Rishu Verma, Renze Lou, Di Liu, Rui Zhang, and Wenpeng Yin. 2024. Large Language Models for Mathematical Reasoning: Progresses and Challenges. *arXiv preprint arXiv:2402.00157*.

Andrew M Bean, Simi Hellsten, Harry Mayne, Jabez Magomere, Ethan A Chi, Ryan Chi, Scott A Hale, and Hannah Rose Kirk. 2024. Lingoly: A benchmark of olympiad-level linguistic reasoning puzzles in low-resource and extinct languages. *arXiv preprint arXiv:2406.06196*.

Gašper Beguš, Maksymilian Dąbkowski, and Ryan Rhodes. 2023. Large linguistic models: Analyzing theoretical linguistic abilities of LLMs. *arXiv preprint arXiv:2305.00948*.

Susan Carey. 2004. Bootstrapping & the origin of concepts. *Daedalus*, 133(1):59–68.

Yuri Chervonyi, Trieu H Trinh, Miroslav Olšák, Xiaomeng Yang, Hoang Nguyen, Marcelo Menegali, Junehyuk Jung, Vikas Verma, Quoc V Le, and Thang Luong. 2025. Gold-medalist Performance in Solving Olympiad Geometry with AlphaGeometry2. *arXiv preprint arXiv:2502.03544*.

Bernard Comrie. 2011. Typology of numeral systems. *Numeral types and changes worldwide. Trends in Linguistics. Studies and monographs*, 118.

Stanislas Dehaene. 2011. *The number sense: How the mind creates mathematics*. Oxford University Press USA.

Ivan Derzhanski and Thomas Payne. 2010. The Linguistics Olympiads: Academic competitions in linguistics for secondary school students. *Linguistics at school: language awareness in primary and secondary education*, pages 213–26.

Ivan Derzhanski and Milena Veneva. 2018. Linguistic Problems on Number Names. In *Proceedings of the Third International Conference on Computational Linguistics in Bulgaria (CLIB 2018)*, pages 169–176.

Meng Fang, Xiangpeng Wan, Fei Lu, Fei Xing, and Kai Zou. 2024. MathOdyssey: Benchmarking Mathematical Problem-solving Skills in Large Language Models using Odyssey Math Data. *arXiv preprint arXiv:2406.18321*.

Lisa Feigenson, Stanislas Dehaene, and Elizabeth Spelke. 2004. Core systems of number. *Trends in cognitive sciences*, 8(7):307–314.

Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, and 1 others. 2025. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv preprint arXiv:2501.12948*.

Harald Hammarström. 2010. Rarities in numeral systems. *Rethinking universals: How rarities affect linguistic theory*, 45:11–53.

Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. 2020. Measuring massive multitask language understanding. *arXiv preprint arXiv:2009.03300*.

James R Hurford. 1987. *Language and number: The emergence of a cognitive system*. B. Blackwell.

Georges Ifrah. 2000. *The Universal History of Numbers*. Harvill London.

Tania Ionin and Ora Matushansky. 2006. The composition of complex cardinals. *Journal of semantics*, 23(4):315–360.